# Didulani Acharige

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## PROFESSIONAL SUMMARY

Machine Learning Research Scientist with 6+ years of experience developing and deploying AI driven solutions for complex real world problems. Skilled in building end-to-end machine learning and deep learning pipelines, from data processing and model development to deployment in operational systems. Demonstrated success in applying deep learning, generative AI, and predictive modeling to complex design and optimization problems, improving model generalization and extrapolation to unseen design spaces. Passionate about translating cutting edge AI research into scalable, impactful solutions through innovation and interdisciplinary collaboration.

## PROJECT BASED EXPERIENCE

**Interface Detection System for Two Photon Lithography System** **[Code - available upon publication]**

- Built an end-to-end deep learning pipeline using a **CNN based regressor** trained on microscope camera images to automatically detect the substrate resin interface position in a two photon lithography (2PL) system, a critical step that determines print origin accuracy and directly impacts 3D nanostructure quality.
- Replaced a manual, operator dependent calibration process with an automated model inference pipeline deployed on the live 2PL system, enabling faster print initialization and reducing positioning errors that compound across nanofabrication workflows, ongoing work focuses on expanding training data diversity across substrate types to improve robustness.

**Generative Tandem Neural Network (GTNN) for Color Splitter Optimization** **[Code] | [Publication]**

- Addressed the single output limitation of conventional **tandem networks** by proposing a novel single input, multiple output generative tandem architecture that produces diverse, high performing design candidates for nanophotonic color splitter devices that redirect different wavelengths of light into spatially separated pixels to reduce absorptive losses in digital camera sensors.
- Incorporated a **Genetic Algorithm** to expand the network's extrapolation capability beyond the training distribution, generating structures with sensitivity and selectivity that exceeded the best values seen in the training dataset.

**Hybrid Deep Learning for Nanophotonic Quantum Emitter Lens Design** **[Publication]**

- Designed a hybrid deep learning pipeline integrating **ResNet50** based **CNNs** with physics driven **adjoint optimization**, directly addressing sensitivity to initial conditions and difficulty integrating multiple design constraints limiting that plague purely data driven or purely physics based approaches.
- Outperformed both standalone deep learning and conventional optimization baselines on quantum emitter lens design tasks.

**ML in Interpolation & Extrapolation for Nanophotonic Inverse Design** **[Publication]**

- Benchmarked **DNN**, **CNN**, **GTNN** and **ML** architectures on forward modeling and inverse design of nanophotonic absorbers evaluating performance across interpolation vs. extrapolation regimes to derive principled guidance on model selection for scientific inverse problems.
- Demonstrated that tandem networks mitigate the non uniqueness problem inherent to inverse scattering, enabling effective training on datasets with ambiguous structure response mappings.

**Deep Learning for high dimensional color splitter structure optimization** **[Code - available upon publication]**

- Benchmarked **GTNN** and **CGANs** for inverse design of high dimensional color splitter structures, evaluating each architecture's ability to generate diverse, high performing design across significantly larger and more complex design space than prior work; systematically explored the extrapolation capability of each approach to assess performance on structures beyond the training distribution.
- Experimentally validated model generated structures by physically fabricating them; fabricated structures matched predicted optical performance, directly confirming the real world reliability of the deep learning pipeline beyond simulation.

**Bird Contamination Removal in Wind Profile Radar Data** **[Code]**

- Trained a **CNN classifier** on 2D radar spectrograms (Time × Doppler Frequency) from MeteoSwiss (Payerne, Switzerland) to detect bird contamination in Radar Wind Profiler signals during migration periods, achieving 86.92% classification accuracy; preprocessing pipeline handled class balancing, grayscale normalization, and rain contaminated sample removal.
- Designed a custom threshold based signal processing filtering algorithm that separates wind signal from bird contamination, reconstructs clean wind profiles, and uses the trained CNN as an automated evaluator, a post filtering false positive rate increase from 19.13% to 54.37% confirms that filtered contaminated samples are successfully transformed to resemble clean wind data.

## PROFESSIONAL EXPERIENCE

**Postdoctoral Research Associate -** Western University, London, ON 2025 – Present

- Led two parallel research thrusts in deep learning for nanophotonic inverse design: benchmarking generative architectures (CGANs, GTNNs) for high dimensional optical structure optimization and developing an end-to-end CNN regression pipeline for interface detection in a 2PL system from model training through live deployment for real time experimental inference.
- Contributed across the full research to fabrication pipeline, from ML driven design of multilayer color splitter structures through physical fabrication and experimental optical characterization, directly validating model predictions against real world performance.

**Post Graduate Research Assistant -** NEM Lab**,** Western University, London, ON 2024 – 2025

- Applied generative AI approaches for inverse design and optimization of nanophotonic color splitter structures, fabricated single and multilayer designs using a 2PL system for experimental validation, and contributed to peer reviewed journal publications.

**Research Assistant -** NEM Lab, Western University, London, ON 2019 – 2024

- Led deep learning research across multiple domains: inverse design of 2D/3D nanophotonic structures (color splitters, quantum nano lenses, solar cells) using CNNs, ResNet50, GTNN and ML approaches and applied deep learning to remove bird contamination from atmospheric wind profile radar data for the Swiss Federal Office of Meteorology.
- Conducted large scale simulations, and applied ensemble modeling techniques to analyze structured and unstructured data across physics and engineering applications.
- Teaching - Graduate TA for Engineering Experimentation, Control Systems, Materials Selection, Energy Conversion

**Assistant Lecturer –** University of Colombo, Sri Lanka 2019 Feb – 2019 June

- Delivered undergraduate lectures, lab sessions, and tutorials while mentoring students in experiments and programming, and collaborating with faculty to improve teaching strategies and learning outcomes.

## EDUCATION

**Ph.D. in Mechanical and Materials Engineering** (**Direct Ph.D.**) Western University, Canada **2019 – 2024**

Thesis – "Deep Learning for inverse design of nanophotonic structures." | CGPA – 91.125/100

**B.Sc. in Engineering Physics Honors** University of Colombo, Sri Lanka **2015 – 2019**

Thesis – "Development of a combinatorial optical property measuring system for classification / quantification of liquids using a machine learning"

## PUBLICATIONS

Google Scholar

- Acharige D , et.al "Machine learning in interpolation and extrapolation for nanophotonic inverse design", *ACS omega*
- Acharige D, et.al. Hybrid Deep Learning for Design of Nanophotonic Lenses, *IOP Nano Express*
- AcharigeD,et.al,"Generative Tandem Neural Network for optimization of photonic structures",*Physica Scripta*
- Acharige D, et.al "Machine learning and artificial neural networks for improved algorithmic design of nanophotonic structures.", Canadian Conference on AI
- Acharige D, et.al "Experimental Validation of Deep Learning Designed Nanophotonic Color Splitters", *Nano Express*, Minor revisions
- Acharige D, et.al "System Level Demonstration of Non Symmetric Nanophotonic Color Splitters for Imaging" (In preparation)

## HONOURS AND AWARDS

- Western Graduate Research Scholarship, Western University, London, Ontario
- MITAC Research Training Award
- Mahapola Higher Education Scholarship, Sri Lanka

## TECHNICAL SKILLS

- **Programming & Scripting**: Python, MATLAB, JavaScript, LaTeX, HTML, CSS
- **ML Frameworks & Libraries**: TensorFlow, Keras, scikit-learn, Pandas, NumPy, OpenCV
- **ML & Deep Learning**: CNN, ResNet, GAN, CGAN, GTNN, PCA, XGBoost, Random Forest
- **Optimization**: Adjoint Optimization, Genetic Algorithm
- **Simulation & Engineering Tools**: Meep (FDTD), SolidWorks, AutoCAD, LabVIEW, Simulink, MultiSim, ModelSim, MATLAB, Inkscape
- **Data & Databases**: MySQL
- **Tools & Version Control**: Git, GitHub